

Synthesis Review: Rates, Extrema, and the Shape of a Graph

Related rates, optimization, and curve analysis

- Problems 1–3 warm up one method each. From problem 4 on they are shuffled, so decide *which* method the problem calls for before you start writing.
- Show the relation or the objective you differentiate. An answer with no setup earns no credit even when the number is right.
- Keep exact values (fractions, π) unless a decimal is requested.

1. A circular oil slick spreads so that its radius grows at $\frac{dr}{dt} = 0.5$ m/s. Find $\frac{dA}{dt}$ at the instant when $r = 20$ m, to two decimal places (square metres per second).

Answer: _____

2. A rectangular garden is fenced on all four sides with 48 m of fence. Let x be one side, so the area is $A(x) = x(24 - x)$.

(a) Find $A'(x)$.

Answer: _____

(b) Find the value of x that maximizes the area.

Answer: _____

3. Let $f(x) = x^3 - 6x^2 + 9x + 2$.

(a) Find $f'(x)$.

Answer: _____

(b) Find the x -values of the critical points.

Answer: _____

(c) Find the x -value of the inflection point.

Answer: _____

4. An open-top box with a square base of edge x centimetres and height h must hold 32000 cm^3 . Its surface area is $S = x^2 + 4xh$.

(a) Use the volume constraint to write S as a function of x alone.

Answer: _____

(b) Find the base edge x that minimizes the surface area.

Answer: _____

5. Let $f(x) = x^4 - 6x^2 + 5$.

(a) Find $f''(x)$.

Answer: _____

(b) Find the x -values of the inflection points.

Answer: _____

(c) Justify whether f has a local maximum, a local minimum, or neither at $x = 0$.

Answer: _____

6. Water runs into an inverted cone at $2 \text{ m}^3/\text{min}$. The cone's radius is always half its water depth h , so $r = \frac{h}{2}$ and $V = \frac{1}{3}\pi r^2 h$.

(a) Write V in terms of h alone.

Answer: _____

(b) Find $\frac{dh}{dt}$ when $h = 4 \text{ m}$, to two decimal places.

Answer: _____

7. A differentiable function g has $g'(x) = (x - 1)^2(x - 4)$.

(a) Find the x -values of the critical points of g .

Answer: _____

(b) Justify, from the sign of g' , what happens to g at each critical point.

Answer: _____

8. Two cars leave the same intersection at the same moment: one travels north at 60 km/h, the other east at 80 km/h. After 2 hours they are 120 km and 160 km from the intersection, so the distance between them is 200 km. How fast is that distance changing at this moment, in kilometres per hour?

Answer: _____

9. A rectangle sits with its base on the x -axis and its two upper corners on the parabola $y = 12 - x^2$. Its area is $A(x) = 24x - 2x^3$ for $x > 0$.

(a) Find $A'(x)$.

Answer: _____

(b) Find every solution of $A'(x) = 0$.

Answer: _____

(c) Find the maximum area.

Answer: _____

10. A spherical balloon is inflated at $\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$, where $V = \frac{4}{3}\pi r^3$ and the surface area is $S = 4\pi r^2$.

- (a) Find $\frac{dr}{dt}$ when $r = 10 \text{ cm}$, to two decimal places.

Answer: _____

- (b) Find $\frac{dS}{dt}$ at that same instant, to two decimal places.

Answer: _____

- (c) Justify whether the surface area grows faster at $r = 10 \text{ cm}$ or at $r = 5 \text{ cm}$.

Answer: _____